Hole Cleaning and Drilling Parameter Guidelines

Lab sample document (synthetic). Not operational guidance.

Why hole cleaning matters

Cuttings that are not removed from the annulus settle on the low side of deviated holes and form beds. Beds raise torque and drag and are a leading cause of pack-off and stuck pipe.

Indicators of poor hole cleaning

- Torque and drag above the modeled trend.
- Overpull when picking up off bottom.
- Rising standpipe pressure at constant flow rate.
- Cuttings volume at the shakers lower than expected for the current ROP.

Parameter guidelines (illustrative)

Hole section	Flow rate (gpm)	RPM	Notes
12-1/4 in. vertical	800 to 950	80 to 120	Pump sweeps every 3 to 4 stands
8-3/4 in. curve	550 to 650	60 to 100	Limit ROP if torque trends up
8-3/4 in. lateral	550 to 650	120 to 160	High RPM aids cuttings transport

Controlled ROP

When hole cleaning indicators appear, reduce ROP until torque and SPP stabilize. Drilling faster than the hole can be cleaned trades short-term footage for a higher chance of NPT.

Circulating before trips and connections

Circulate bottoms up at least once before tripping out of a deviated section. In high-risk intervals, circulate and rotate for the time of one bottoms up before each connection.